

Soumaya Lamrhari

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[Google Scholar Profile](#) | [LinkedIn Profile](#)

EDUCATION

Mohammed V University , School of Computer Science and Systems Analysis (ENSIAS) Ph.D. in Computer Science	Rabat, Morocco Dec 2015 - Nov 2022
- Cumulative GPA: 4.0/4.0 - Dissertation: Social CRM and Business Intelligence Analytics: Approaches for Customer Retention - Grade: Very honorable	
University of Ibn Tofail , School of Engineering and Applied Science Master in Information Systems Security	Kenitra, Morocco July 2014
University of Mohammed V , Faculty of Science Bachelor in Computer Science	Rabat, Morocco July 2012

PROFESSIONAL EXPERIENCE

Mohammed V University, ENSIAS <i>Graduate Research Assistant</i>	Rabat, Morocco Dec 2015 - Nov 2022
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Project 1: Predicting Customer Conversion

- Built an XGBoost model to predict 30-day customer conversion using CRM usage data.
- Handled class imbalance and optimized precision-recall threshold for prospect prioritization.
- Identified key behavioral features and presented results with PR curve and feature importance.

Project 2: Churn Prediction

- Built a churn prediction model using historical CRM data to identify customers likely to leave within 30 days.
- Performed EDA, feature engineering (e.g. tenure, interaction frequency) and trained an XGBoost model in Python.
- Evaluated with AUC and precision/recall and used feature importance/SHAP to highlight key churn drivers for retention actions.

Project 3: Decision Support Framework Using Text Mining and Sentiment Analysis

- Developed a framework to transform customer feedback into actionable insights using text mining (LDA) and the Fuzzy-Kano model.
- Measured customer satisfaction by extracting product aspects and mapping requirements to a dynamic SWOT matrix.
- Helped businesses better understand customer needs and adapt product offerings.

Project 4: Customer Retention Improvement with Machine Learning

- Built a Random Forest model to classify customers as prospects, satisfied, or unsatisfied.
- Enabled businesses to improve customer targeting and retention strategies by identifying key customer segments.
- Helped optimize marketing efforts and boost profitability.

Project 5: Customer Retention Optimization

- Developed a linear optimization model to maximize customer satisfaction and conversion rates.
- Enhanced customer engagement using social network analysis and eWOM (electronic Word of Mouth).
- Proposed cost-effective strategies to improve customer retention and loyalty.

PUBLICATIONS

Journal Articles

- Lamrhari, S., Elghazi, H., Oubrich, M., & El Faker, A. (2022). A social CRM analytic framework for improving customer retention, acquisition, and conversion. *Technological Forecasting and Social Change*, 174, 121275.
- Lamrhari, S., Elghazi, H., & El Faker, A. (2019). Business intelligence using the fuzzy-Kano model. *Journal of Intelligence Studies in Business*, 9(2).

Conference Papers

- Lamrhari, S., Elghazi, H., & El Faker, A. (2020, November). Random Forest-based Approach for Classifying Customers in Social CRM. In *2020 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)* (pp. 1-6). IEEE.

- Lamrhari, S., Elghazi, H., Tigani, S., El Faker, A., & Sadiki, T. (2018, March). Enhancing Social Network Communication through Dynamic Clustering Balance. In Proceedings of the 2nd Mediterranean Conference on Pattern Recognition and Artificial Intelligence (pp. 63-69).
- Lamrhari, S., Elghazi, H., Sadiki, T., & El Faker, A. (2016, May). A profile-based Big data architecture for agricultural context. In 2016 International Conference on Electrical and Information Technologies (ICEIT) (pp. 22-27). IEEE.

LEADERSHIP EXPERIENCE

National Institute of Post and Telecommunication (INPT)

Rabat, Morocco

Teaching Assistant, Unified Modeling Language (UML)

Jan 2018 - Jun 2018

- Led graduate students through UML concepts and their applications in software design and system architecture.
- Mentored students in creating class, use case, sequence, and activity diagrams to strengthen object-oriented design skills.
- Developed instructional materials and provided detailed feedback to enhance learning outcomes and foster critical thinking.

Mohammed V University, ENSIAS

Rabat, Morocco

Teaching Assistant, Sentiment Analysis Projects

Feb 2018 - Jul 2018

- Mentored graduate students on sentiment analysis projects using NLP techniques, text preprocessing, and machine learning models.
- Provided guidance on text classification, improving model accuracy, and presenting insights from sentiment data.
- Supported students in implementing practical solutions for real-world applications of sentiment analysis.

CERTIFICATIONS & PERSONAL DEVELOPMENT

Professional Development Certificate

Nov 2021- Dec 2021

- Completed 75 hours of online training on essential soft skills, including communication, teamwork, problem-solving, adaptability, and time management.

Big Data and Analytics Summer School (University of Essex)

Colchester, UK

Jul 2017- Aug 2017

- Completed a two-week program focused on big data management, AI, and statistical methods.
- Explored applications in business, finance, and bioinformatics.
- Participated in lab-based and theoretical courses delivered by academic and industry experts.